

Exercise Sheet 5

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Due date: 13:00, May 14th.

Submissions that are late will receive no credit.

You should try to solve all of the exercises below — each problem is worth 10 points. Make sure to clearly justify your answers, defining everything precisely and stating any results you are using. You should then submit your solutions in Gradescope on NTU COOL, either as a typed PDF or a clear scan of your handwritten answers, and should indicate where in the file the solutions to each exercise are. If you have difficulties with Gradescope, please send your solutions by e-mail.

Exercise 1 X is a random variable whose cumulative distribution function F_X is given by

$$F_X(x) = \begin{cases} 0 & x \leq 0, \\ x^2 & 0 \leq x \leq 1, \\ 1 & 1 \leq x. \end{cases}$$

Determine the expectation and variance of X .

Solution: Since

$$f_X(x) = F'_X(x) = \begin{cases} 0, & \text{if } x \notin [0, 1]; \\ 2x, & \text{if } x \in [0, 1]. \end{cases}$$

We have

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} t f_X(t) dt = \int_0^1 t \cdot 2t dt = \frac{2}{3}.$$

Also,

$$\mathbb{E}[X^2] = \int_{-\infty}^{\infty} t^2 f_X(t) dt = \int_0^1 t^2 \cdot 2t dt = \frac{1}{2}.$$

Thus,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}.$$

Exercise 2 Two brothers, Alpha and Beta, have a 30cm long chocolate bar to share. They break the bar at a uniformly random point, and then the older brother, Alpha, takes the larger piece, while Beta gets the shorter piece. What is the expectation of the length of Alpha's piece of chocolate?

Solution: Suppose when we break the chocolate bar, we put it on a platform, and define X to be the length (cm) from the breakpoint to the left of chocolate bar. Then, we know $X \sim \text{Unif}([0, 30])$. Since Alpha will take the larger piece, so we define the length of chocolate bar that Alpha gets to be

$$Y \sim \max\{X, 30 - X\},$$

which is because the broken chocolate bar has 2 portions, one of length X , while the other one of length $30 - X$, and Alpha will take the larger piece. Note that $Y \geq 15$ since if $X \leq 15$, $30 - X \geq 15$, otherwise $X \geq 15$ also gives $Y \geq 15$. Now for $t \in [15, 30]$, we know

$$\begin{aligned} F_Y(t) &= \mathbb{P}(\max\{X, 30 - X\} \leq t) = \mathbb{P}(X \leq t, 30 - X \leq t) \\ &= \mathbb{P}(30 - t \leq X \leq t) = \int_{30-t}^t f_X(x) dx \\ &= \int_{30-t}^t \frac{1}{30} dx = \frac{1}{30} (t - (30 - t)) = \frac{1}{15} t - 1. \end{aligned}$$

Hence, we have

$$f_Y(t) = \begin{cases} F'_Y(t) = \frac{1}{15}, & \text{if } 15 \leq t \leq 30; \\ 0, & \text{if } t \notin [15, 30]. \end{cases}$$

Thus,

$$\mathbb{E}[Y] = \int_{15}^{30} t \cdot \frac{1}{15} dt = \left[\frac{t^2}{30} \right]_{15}^{30} = 22.5.$$

Exercise 3 X and Y are two independent exponential random variables with the rate λ . Let $Z = \max\{X, Y\}$.

- (a) What is the probability density function of Z ?
- (b) What is the expectation of Z ?

Solution:

- (a) Note that

$$\begin{aligned} F_Z(t) &= \mathbb{P}(Z \leq t) = \mathbb{P}(\max\{X, Y\} \leq t) = \mathbb{P}(X \leq t, Y \leq t) \\ &= \mathbb{P}(X \leq t)\mathbb{P}(Y \leq t) \quad [\text{independence}] \\ &= F_X(t)F_Y(t) = \begin{cases} (1 - e^{-\lambda t})^2, & \text{if } t \geq 0; \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Thus,

$$f_Z(t) = F'_Z(t) = \begin{cases} 2(1 - e^{-\lambda t}) \cdot (\lambda e^{-\lambda t}), & \text{if } t \geq 0; \\ 0, & \text{if } t < 0. \end{cases}$$

(b)

$$\begin{aligned}\mathbb{E}[Z] &= \int_{-\infty}^{\infty} t f_Z(t) dt = \int_0^{\infty} 2\lambda t(1 - e^{-\lambda t}) dt = 2\lambda \left(\int_0^{\infty} t e^{-\lambda t} dt - \int_0^{\infty} t e^{-2\lambda t} dt \right) \\ &= 2\lambda \left(- \int_{-\infty}^0 \frac{u}{\lambda^2} e^u du + \int_{-\infty}^0 \frac{u}{4\lambda^2} e^u du \right) = \left(-\frac{2}{\lambda} + \frac{1}{2\lambda} \right) \int_{-\infty}^0 u e^u du \\ &= \left(-\frac{3}{2\lambda} \right) \int_{-\infty}^0 u e^u du = \frac{3}{2\lambda}.\end{aligned}$$

Exercise 4 There are two tour bus routes that stop at the same bus stop - a red bus and a green bus. The waiting times for the buses are all independent, with the times between consecutive red buses exponentially distributed with rate λ_1 , while the times between consecutive green buses are exponentially distributed with rate λ_2 .

- (a) A tourist arrives at the bus stop. What is the probability that the first bus to arrive will be a red bus?
- (b) The tourist has already bought a ticket for the green bus. What is the expected number of red buses that will arrive before the first green bus?

Solution:

- (a) Suppose X is the time of first red bus to arrive, and Y is the time of the first green bus to arrive, then $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$ and X, Y are independent. Hence,

$$\begin{aligned}\mathbb{P}(\text{first bus is red}) &= \mathbb{P}(X < Y) = \int_0^{\infty} \int_u^{\infty} f_{X_1, X_2}(u, v) dv du \\ &= \int_0^{\infty} \int_u^{\infty} f_{X_1}(u) f_{X_2}(v) dv du \quad [\text{independence}] \\ &= \int_0^{\infty} \int_u^{\infty} \lambda_1 e^{-\lambda_1 u} \cdot \lambda_2 e^{-\lambda_2 v} dv du \\ &= \int_0^{\infty} \lambda_1 e^{-\lambda_1 u} \left(\int_u^{\infty} \lambda_2 e^{-\lambda_2 v} dv \right) du \\ &= \int_0^{\infty} \lambda_1 e^{-\lambda_1 u - \lambda_2 u} du \\ &= \frac{\lambda_1}{\lambda_1 + \lambda_2}.\end{aligned}$$

- (b) We model red and green buses as two independent Poisson processes with rates λ_1 and λ_2 , respectively. Let $(R(t))_{t \geq 0}$ and $(G(t))_{t \geq 0}$ be their counting processes. Let S_i denote the arrival time of the i -th red bus, and let Y be the arrival time of the first green bus:

$$Y = \inf\{t \geq 0 : G(t) \geq 1\}.$$

Define

$$N = \max\{i \geq 0 : S_i < Y\},$$

the number of red arrivals before the first green arrival.

Step 1: distribution of event ordering. At any time, the next red arrival occurs after an exponential waiting time with rate λ_1 , and the next green arrival occurs after an independent exponential waiting time with rate λ_2 . Hence, for independent $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$, we have shown that

$$\mathbb{P}(X < Y) = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

in (a), and

$$\mathbb{P}(Y < X) = 1 - \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$

Step 2: memoryless property and stopping times. Let $Y \sim \text{Exp}(\lambda_2)$ and let $T \geq 0$ be a continuous random variable independent of Y (in particular, T may be a stopping time determined by the red process such as S_i). Then for any $s \geq 0$,

$$\mathbb{P}(Y - T > s \mid Y > T) = \frac{\mathbb{P}(Y > T + s)}{\mathbb{P}(Y > T)}.$$

We compute numerator and denominator by conditioning on T :

$$\begin{aligned} \mathbb{P}(Y > T + s) &= \int_0^\infty \mathbb{P}(Y > t + s) f_T(t) \, dt = \int_0^\infty e^{-\lambda_2(t+s)} f_T(t) \, dt \\ &= e^{-\lambda_2 s} \int_0^\infty e^{-\lambda_2 t} f_T(t) \, dt, \end{aligned}$$

and

$$\mathbb{P}(Y > T) = \int_0^\infty e^{-\lambda_2 t} f_T(t) \, dt.$$

Hence,

$$\mathbb{P}(Y - T > s \mid Y > T) = e^{-\lambda_2 s}.$$

Therefore,

$$Y - T \mid (Y > T) \sim \text{Exp}(\lambda_2).$$

Applying this with $T = S_i$, and using that S_i depends only on the red process and is independent of the green process, we obtain

$$Y - S_i \mid (S_i < Y) \sim \text{Exp}(\lambda_2).$$

Step 3: regeneration. At each red arrival time before Y , the residual green waiting time has the same exponential distribution and is independent of the past. Moreover, by independence of increments of Poisson processes, the future evolution after any such time has the same law as the original system. Hence after each red arrival (before the first green), the process probabilistically restarts.

Thus, the sequence of observed bus colors forms a sequence of independent trials where each event is red with probability

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} \quad \text{and green with probability} \quad \frac{\lambda_2}{\lambda_1 + \lambda_2}.$$

The process stops at the first green arrival.

Step 4: distribution of N . For $k \geq 0$, the event $\{N = k\}$ means that the first k arrivals are red and the $(k + 1)$ -th is green. Hence

$$\mathbb{P}(N = k) = \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right).$$

Step 5: expectation via integral. We compute

$$\mathbb{E}[N] = \sum_{k=0}^{\infty} k \mathbb{P}(N = k) = \sum_{k=0}^{\infty} k \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right).$$

Let $p = \frac{\lambda_2}{\lambda_1 + \lambda_2}$ and $q = \frac{\lambda_1}{\lambda_1 + \lambda_2}$. Then

$$\mathbb{E}[N] = p \sum_{k=0}^{\infty} k q^k.$$

Using the standard identity (derived from differentiating $\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$ for $|q| < 1$),

$$\sum_{k=0}^{\infty} k q^k = \frac{q}{(1-q)^2},$$

we obtain

$$\mathbb{E}[N] = p \cdot \frac{q}{(1-q)^2}.$$

Substituting $1 - q = p$ gives

$$\mathbb{E}[N] = \frac{q}{p} = \frac{\lambda_1}{\lambda_2}.$$

Conclusion.

$$\boxed{\mathbb{E}[N] = \frac{\lambda_1}{\lambda_2}.$$